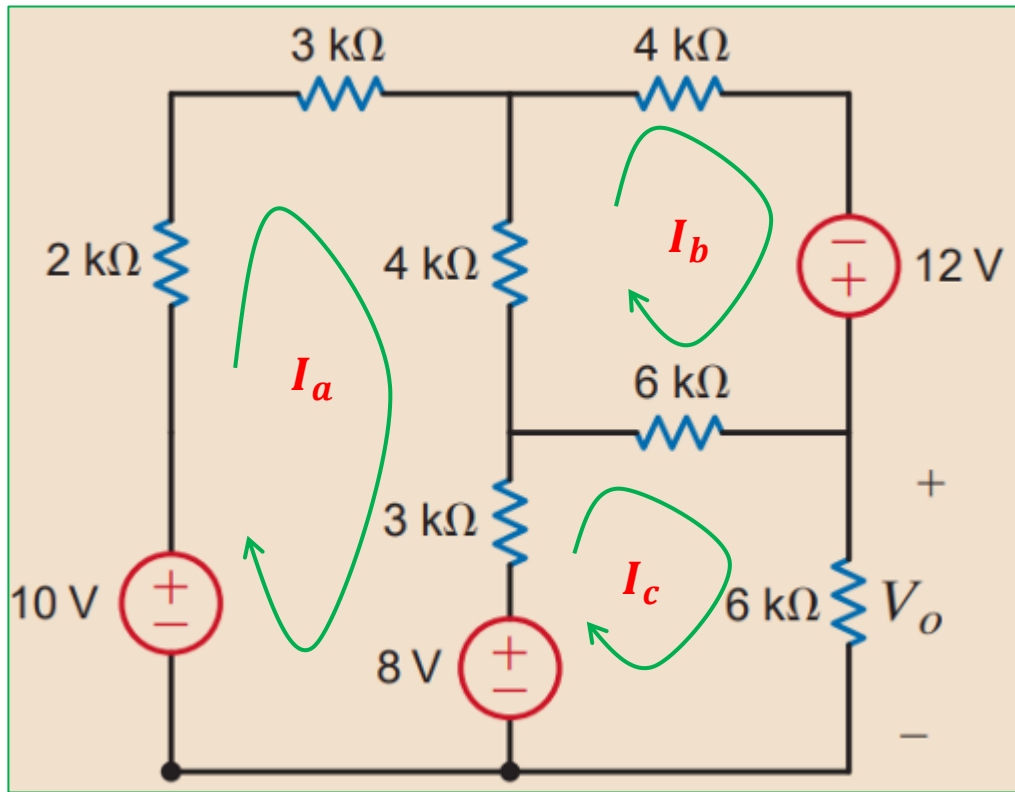


Steps to write Mesh equations

1. Count the number of Mesh (independent loops). Let's say number of Mesh is = n . [$n = 1, 2, 3 \dots$]
2. Assign a current for each mesh. Total number of assigned currents = n .
3. Obtain n numbers of equations in terms of the assigned currents.
4. If there is **NO** current source in the circuit, obtain n numbers of equations applying KVL at each loops.
5. If there are current sources in the circuit, use Supermesh to obtain the equations.



If there is **NO** current source in the circuit

1. Number of Mesh: 3
2. Assign mesh currents: I_a, I_b and I_c
3. Number of equations: 3 to be expressed in terms I_a, I_b and I_c .
4. Since, no current sources in the circuit Apply only KVL in each loop to obtain the equations

KVL at Loop a:

$$+10 - 2I_a - 3I_a - 4(I_a - I_b) - 3(I_a - I_c) - 8 = 0$$

KVL at Loop b:

$$-4(I_b - I_a) - 4I_b + 12 - 6(I_b - I_c) = 0$$

KVL at Loop c:

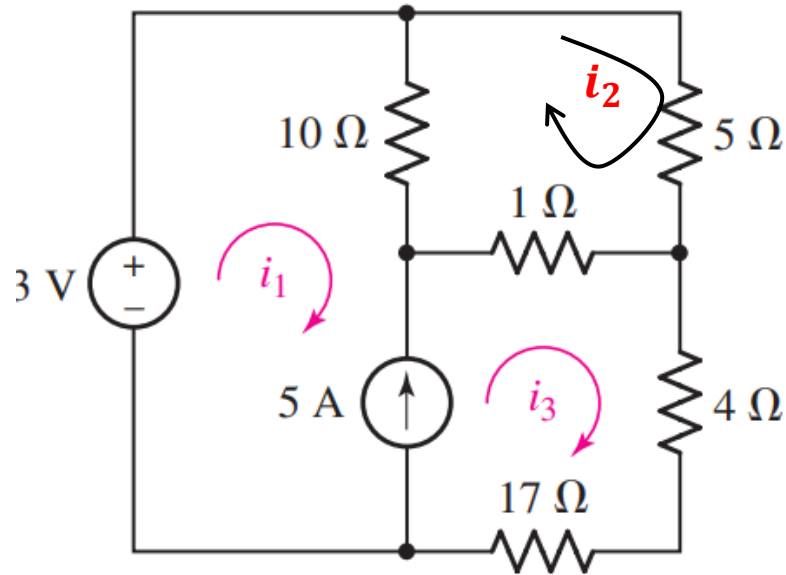
$$8 - 3(I_c - I_a) - 6(I_c - I_b) - 6I_c = 0$$

Solve Three equations to determine I_a, I_b and I_c

Write the supermesh equation in the circuit given below:

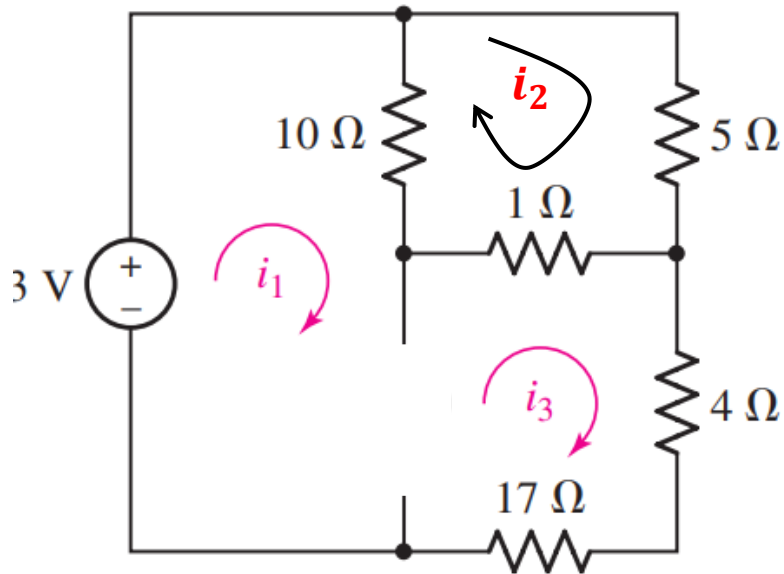
1. Number of Mesh: 3
2. Assign mesh currents: i_1, i_2 and i_3
3. Number of equations: 3 to be expressed in terms i_1, i_2 and i_3

1. Obtain one equation relating i_1 and i_3 : $i_3 - i_1 = 5$
2. Obtain 2nd equation applying KVL in loop 2: $-10(i_2 - i_1) - 5i_2 - 1(i_2 - i_3) = 0$



3. 3rd equation:

1. Redraw the circuit opening the current sources to obtain super mesh.
2. Apply KVL around the supermesh.



KVL around the supermesh

$$3 - 10(i_1 - i_2) - 1(i_3 - i_2) - 4i_3 - 17i_3 = 0$$